Long-CoT Reinforcement Learning and Localized Policy Optimization

Deep Dive into Kimi K3 — Chapter 19

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Foundation Books Project

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Goal: compare attempts before assigning credit, separate reward from restoring movement, choose and preserve a reduction, and report staleness as realized displacement — with every Kimi claim kept in its evidence class.

Roadmap

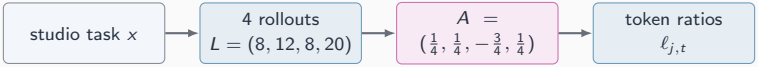
Relative credit and hidden movement

The localized update

Reductions, rows, and staleness

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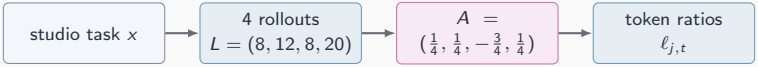
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- Outcomes $(1, 1, 0, 1)$ center to relative credit; direct and recovered success share $\frac{1}{4}$ without valuing the initial error.

[DERIVED: chapter studio fixtures]

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- Outcomes $(1, 1, 0, 1)$ center to relative credit; direct and recovered success share $\frac{1}{4}$ without valuing the initial error.
- Exact factorization: $\ell_j^{\text{seq}} = \sum_t \ell_{j,t}$. Yet $(0.3, -0.3)$ sums to zero with local-square sum 0.18 — the sequence hides movement.

[DERIVED: chapter studio fixtures]

The localized update

Reward within a band, restoration everywhere

$$\mathcal{S}_{\text{local}} = \frac{1}{N}\sum_{j,t}\Big[A_j\phi_{a,b}(\ell_{j,t}) - \lambda_{\text{reg}}\ell_{j,t}^2\Big], \qquad g_{j,t} = \big(I_{j,t}^{\text{loc}} A_j e^{\ell_{j,t}} - 2\lambda_{\text{reg}}\ell_{j,t}\big)\nabla_\theta \log \pi_\theta(a_{j,t} \mid h_{j,t}).$$

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- $A = 0.25$, $\ell = 0.3$: reward coefficient 0, restoring -0.06. Unlike PPO clipping, the gate is independent of the advantage sign.
- One real step on a binary policy: $(0.3, 0.7) \to (0.3227, 0.6773)$ — coefficients must multiply their own score vectors.

Reductions, rows, and staleness

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Keep outcome signal, local control, reduction, and collection apart. Next: Chapter 20's Sampling, Curricula, and Partial Rollouts.